

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month we look at hypothetical particles, Cryopegs, RNA-based therapies, Breakthrough Listen and more. We also have a fun slime DIY section. We also delve into an emerging science known as optogenetics which can pave way for new treatments for multiple disorders.



Musk announces Neuralink plans

Elon Musk has revealed more details about Neuralink, a company that intends to create brain computer interfaces, with the ultimate goal of merging humanity with AI. The company intends to start with neural implants that will help paralysed people control devices such as computers and smartphones. <https://digit.in/nrlnk>

WHAT'S NEW

Moon rocks continue to drive science

On July 20, 1969, the Eagle landed on the Moon, with two astronauts on board. One primary objective was to return samples of Moon rocks to Earth. The astronauts, Buzz Aldrin and Neil Armstrong, collected these samples using hammers, tongs and scoops. The subsequent Apollo missions were also used to collect samples. In all, the missions collected over 380 kg of lunar samples. These samples were not studied immediately. NASA stored away some of the samples, in vacuum-sealed containers. For fifty years, these samples were never exposed to the atmosphere. The idea was to keep the samples stored till there were more advanced technologies to study them.



NASA is now releasing some of these stored samples from the Apollo 17 mission to educational and research institutions. As far as

researchers are concerned, it is as if there has been a new sample return mission to the Moon. The science based on the Apollo mission continues, even after half a century. There are still questions to be answered, for example researchers don't know if the Moon is still geologically active. <https://digit.in/moonrocks>

Acoustic levitation allows for studying chemistry

Sound waves propagate through air by creating high pressure and low pressure areas. An array of transducers can control these areas, using ultrasonic sound waves. However, bats in the area would be disturbed by the noise. By controlling the alternating high and low pressure areas, objects can be trapped or levitated. Such devices can be made at home, using cheap components from a hardware store.



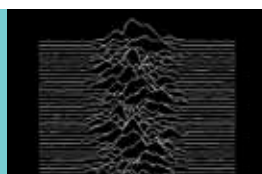
In a new study, Caltech scientist Jack Beauchamp described the use of acoustic levitation to study the chemistry of skin cancer drugs. Beauchamp calls the approach "lab-in-a-drop". A 1mm water droplet was coated with lipids. When introduced to

the suspended water droplet, the lipids rose to the surface to form a thin membrane around the drop. The drug Temoporfin was introduced to the suspended droplet. When excited with a red laser light, Temoporfin destroys molecules it comes in contact with. In potential skin cancer treatments, a laser can be shone on the affected areas, to selectively eliminate the cancerous tissue. <https://digit.in/auclev>



How mosquito brains work

Carbon dioxide exhaled by humans and other animals are a trigger, which activates the visual centers of the mosquito brain. The mosquito begins to scan the surroundings for shapes that could match the target, or the source of carbon dioxide, and begins flying towards it. <https://digit.in/mosbrn>



The "Unknown Pleasures" Pulsar

Astronomers observed PSR B1919+21 on the anniversary of the Joy Division album. When first released, little was known about pulsars. Changes in the brightness of radio flashes were thought to be chaotic. Scientists can now see periodic waxing and waning of the brightness of some pulsars. <https://digit.in/jdpsr>



Hubble discovers erratic black hole

The Hubble Space telescope found a disk around a black hole that shouldn't be where it is. The disk is very close to the supermassive black hole, and lets scientists take a peek at the interesting physics in the accretion disk, where particles travel at relativistic speeds. <https://digit.in/hblbh>